Data Ingestion from the RDS to HDFS using Sqoop

I. Sqoop Import command used for importing table from RDS to HDFS:

```
sqoop import \
--connect jdbc:mysql://upgraddetest.cyaielc9bmnf.us-east-1.rds.amazonaws.com/testdatabase \
--username student \
--password STUDENT123 \
--table SRC_ATM_TRANS \
--target-dir /user/hadoop/atm_raw \
--as-textfile \
-m 1
```

Explanation of the Sqoop Parameters:

--connect, username, password, and table:
Credentials and source table provided by the assignment.

--target-dir /user/hadoop/atm_raw
Specifies the HDFS directory where the imported data will be stored.

--as-textfile
Imports the data in plain text forma for schema-based reading later using PySpark.

-m 1
Forces Sqoop to use a single mapper. This option was required because the source table does not have a primary key that could be used for parallel splitting.

II. Command used to see the list of imported data in HDFS:

```
hdfs dfs -ls /user/hadoop/atm_raw
hdfs dfs -head /user/hadoop/atm_raw/part-m-00000
```

III. Screenshot of the imported data:

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```
2026-04-08 07:24:51,083 INFO mapreduce.ImportJobBase: Transferred 506.6059 MB in 53.9211 seconds (9.3953 MB/sec)
2026-04-08 07:24:51,086 INFO mapreduce.ImportJobBase: Retrieved 2468572 records.
[hadoop@ip-172-31-21-134 ~]$ hdfs dfs -ls /user/hadoop/atm_raw
Found 2 items
-rw-r--r--   1 hadoop hdfsadmingroup          0 2026-04-08 07:24 /user/hadoop/atm_raw/_SUCCESS
-rw-r--r--   1 hadoop hdfsadmingroup  531214815 2026-04-08 07:24 /user/hadoop/atm_raw/part-m-00000
[hadoop@ip-172-31-21-134 ~]$ hdfs dfs -head /user/hadoop/atm_raw/part-m-00000
2017,January,1,Sunday,0,Active,1,NCR,NÃƒÂ¦stved,Farimagsvej,8,4700,55.233,11.763,DKK,MasterCard,5643,Withdrawal,,,55.230,11.761,2616038,Naestved,281.150,1014,87,7,260,0.21
5,92,500,Rain,light rain
2017,January,1,Sunday,0,Inactive,2,NCR,Vejgaard,Hadsundvej,20,9000,57.043,9.950,DKK,MasterCard,1764,Withdrawal,,,57.048,9.935,2616235,NÃƒÂ¸rresundby,280.640,1020,93,9,250,
0.590,92,500,Rain,light rain
2017,January,1,Sunday,0,Inactive,2,NCR,Vejgaard,Hadsundvej,20,9000,57.043,9.950,DKK,VISA,1891,Withdrawal,,,57.048,9.935,2616235,NÃƒÂ¸rresundby,280.640,1020,93,9,250,0.590,
92,500,Rain,light rain
2017,January,1,Sunday,0,Inactive,3,NCR,Ikast,RÃƒÂ¥dhusstrÃƒÂ¦det,12,7430,56.139,9.154,DKK,VISA,4166,Withdrawal,,,56.139,9.158,2619426,Ikast,281.150,1011,100,6,240,0.000,75
,300,Drizzle,light intensity drizzle
2017,January,1,Sunday,0,Active,4,NCR,Svogerslev,BrÃƒÂ¸nsager,1,4000,55.634,12.018,DKK,MasterCard,5153,Withdrawal,,,55.642,12.080,2614481,Roskilde,280.610,1014,87,7,260,0.0
00,88,701,Mist,mist
2017,J[hadoop@ip-172-31-21-134 ~]$
```

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